

A Riemannian Geometric Description of Characteristic Classes

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Abstract

This report summarizes the main concepts and results presented in Appendix C of Milnor and Stasheff's *Characteristic Classes*, which develops the Chern–Weil description of characteristic classes in terms of curvature forms. We cover connections on complex and real vector bundles, the curvature tensor, invariant polynomials, the fundamental lemma, and the expressions of Chern classes, Pontrjagin classes, and the Euler class via curvature.

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1 Connections on Complex Vector Bundles

Throughout this report, M denotes a smooth manifold. For a complex vector bundle ξ over M , we write

$$\tau_{\mathbb{C}}^* = \text{Hom}_{\mathbb{R}}(\tau, \mathbb{C})$$

for the complexified dual tangent bundle of M . We denote by $C^\infty(\xi)$ the vector space of smooth sections of ξ , and similarly for other bundles.

Definition 1.1 (Connection). A *connection* on a smooth complex n -plane bundle ξ is a $\mathbb{C}$ -linear mapping

$$\nabla : C^\infty(\xi) \longrightarrow C^\infty(\tau_{\mathbb{C}}^* \otimes \xi)$$

which satisfies the Leibniz formula

$$\nabla(fs) = df \otimes s + f \nabla s$$

for every $s \in C^\infty(\xi)$ and every smooth function $f : M \rightarrow \mathbb{C}$. The image ∇s is called the *covariant derivative* of s .

Remark 1.2. An important property of a connection is that it is a *local operator*: if a section s vanishes on an open set U , then ∇s also vanishes on U . Consequently, it makes sense to restrict a connection to an open subset, and a global connection is uniquely determined by its restrictions to an open cover.

Lemma 1.3 (Local description). *Let $U \subset M$ be an open set small enough so that $\xi|_U$ is trivial, and let $s_1, \dots, s_n$ be a basis for the sections of $\xi|_U$. Then a connection ∇ on $\xi|_U$ is uniquely determined by $\nabla s_1, \dots, \nabla s_n$, which can be completely arbitrary smooth sections of $\tau_{\mathbb{C}}^* \otimes \xi|_U$. Each ∇s_i can be written uniquely as*

$$\nabla s_i = \sum_j \omega_{ij} \otimes s_j,$$

where $[\omega_{ij}]$ can be an arbitrary $n \times n$ matrix of smooth complex 1-forms on U . The matrix $[\omega_{ij}]$ is called the *connection matrix with respect to the basis $\{s_i\}$* .

Remark 1.4. The collection of all connections on ζ does not have any natural vector space structure. Note however that if ∇_1 and ∇_2 are two connections on ζ , and g is a smooth complex valued function on M , then the linear combination $g\nabla_1 + (1 - g)\nabla_2$ is again a well defined connection on ζ .

Lemma 1.5 (Existence of connections). Every smooth complex vector bundle with paracompact base space possesses a connection.

Lemma 1.6 (Induced bundles). Let $g : M' \rightarrow M$ be a smooth map and $\xi' = g^*\xi$ the induced vector bundle over M' . Note that there is a canonical $C^\infty(M, \mathbb{C})$ -linear mapping $g^* : C^\infty(\xi) \rightarrow C^\infty(\xi')$. Also, any 1-form on M pulls back to a 1-form on M' , so there is a canonical $C^\infty(M, \mathbb{C})$ -linear mapping $g^* : C^\infty(\tau_{\mathbb{C}}^* \otimes \xi) \rightarrow C^\infty(\tau_{\mathbb{C}}^* \otimes \xi')$. To each connection ∇ on ξ there corresponds one and only one connection $\nabla' = g^*\nabla$ on ξ' so that the following diagram is commutative:

$$\begin{array}{ccc} C^\infty(\xi) & \xrightarrow{\nabla} & C^\infty(\tau_{\mathbb{C}}^* \otimes \xi) \\ g^* \downarrow & & \downarrow g^* \\ C^\infty(\xi') & \xrightarrow{\nabla'} & C^\infty(\tau_{\mathbb{C}}^* \otimes \xi') \end{array}$$

2 The Curvature Tensor

Given a connection ∇ on ξ , one can extend it to a map on $C^\infty(\tau_{\mathbb{C}}^* \otimes \xi)$ by combining it with exterior differentiation.

Lemma 2.1. Given ∇ , there is one and only one $\mathbb{C}$ -linear mapping

$$\hat{\nabla} : C^\infty(\tau_{\mathbb{C}}^* \otimes \xi) \longrightarrow C^\infty(\wedge^2 \tau_{\mathbb{C}}^* \otimes \xi)$$

which satisfies the Leibniz formula

$$\hat{\nabla}(\alpha \otimes s) = d\alpha \otimes s - \alpha \wedge \nabla s$$

for every 1-form α and every section $s \in C^\infty(\xi)$.

Now consider the composition $K = \hat{\nabla} \circ \nabla$:

$$C^\infty(\xi) \xrightarrow{\nabla} C^\infty(\tau_{\mathbb{C}}^* \otimes \xi) \xrightarrow{\hat{\nabla}} C^\infty(\wedge^2 \tau_{\mathbb{C}}^* \otimes \xi).$$

Lemma 2.2. The value of $K(s)$ at a point $x \in M$ depends only on $s(x)$, not on the values of s at other points. Hence the correspondence $s(x) \mapsto K(s)(x)$ defines a smooth section of the complex vector bundle $\text{Hom}(\xi, \wedge^2 \tau_{\mathbb{C}}^* \otimes \xi) \cong \wedge^2 \tau_{\mathbb{C}}^* \otimes \text{Hom}(\xi, \xi)$.

Definition 2.3 (Curvature tensor). The section $K = K_\nabla$ of $\wedge^2 \tau_{\mathbb{C}}^* \otimes \text{Hom}(\xi, \xi)$ is called the *curvature tensor* of the connection ∇ .

In terms of a local basis $s_1, \dots, s_n$ with $\nabla s_i = \sum_j \omega_{ij} \otimes s_j$, the curvature is given locally by

$$K(s_i) = \sum_j \Omega_{ij} \otimes s_j, \quad \Omega_{ij} = d\omega_{ij} - \sum_k \omega_{ik} \wedge \omega_{kj}.$$

In matrix notation, with $\omega = [\omega_{ij}]$ and $\Omega = [\Omega_{ij}]$, we have the fundamental formula

$$\Omega = d\omega - \omega \wedge \omega. \tag{1}$$

The matrix Ω is called the *curvature matrix*.

A connection is called *flat* if its curvature tensor is identically zero. A fundamental theorem (not proved here) asserts that $K = 0$ if and only if, in a neighborhood of each point, there exists a basis of sections with zero covariant derivatives.

3 Characteristic Classes from Curvature

3.1 Invariant Polynomials

Let $M_n(\mathbb{C})$ be the algebra of $n \times n$ complex matrices.

Definition 3.1 (Invariant polynomial). An *invariant polynomial* on $M_n(\mathbb{C})$ is a function $P : M_n(\mathbb{C}) \rightarrow \mathbb{C}$ which can be expressed as a complex polynomial in the entries of the matrix and satisfies

$$P(XY) = P(YX),$$

or equivalently

$$P(TXT^{-1}) = P(X)$$

for every non-singular matrix T .

For any square matrix A , let $\sigma_k(A)$ denote the k -th elementary symmetric function of the eigenvalues of A , so that

$$\det(I + tA) = 1 + t\sigma_1(A) + \cdots + t^n\sigma_n(A).$$

Lemma 3.2. Any invariant polynomial on $M_n(\mathbb{C})$ can be expressed as a polynomial function of $\sigma_1, \dots, \sigma_n$.

3.2 The Fundamental Lemma

Given an invariant polynomial P , one can evaluate it on the curvature matrix Ω . Since the entries of Ω are 2-forms (which commute because they are of even degree), this evaluation makes sense and yields a differential form $P(K)$ on the base space M . A change of basis replaces Ω by $T\Omega T^{-1}$, and invariance of P guarantees that $P(K)$ is globally well defined.

Remark 3.3. If P is homogeneous of degree r , then $P(K)$ is an exterior form of degree $2r$. One can also use invariant formal power series; since $P_r(K) = 0$ for $2r > \dim M$, the sum is finite.

Theorem 3.4 (Fundamental Lemma). For any invariant polynomial (or invariant formal power series) P , the exterior form $P(K)$ is closed; that is,

$$dP(K) = 0.$$

Sketch of proof: The proof relies on the *Bianchi identity*

$$d\Omega = \omega \wedge \Omega - \Omega \wedge \omega, \tag{2}$$

obtained by differentiating (1), together with the fact that the matrix of formal partial derivatives of an invariant polynomial commutes with the original matrix.

Corollary 3.5. The cohomology class $(P(K))$ is independent of the connection ∇ .

Sketch of proof: The proof considers the pullback of the bundle to $M \times \mathbb{R}$, constructs a linear combination $\nabla = t\nabla_1 + (1 - t)\nabla_0$ of two connections, and uses the homotopy invariance of de Rham cohomology.

Because the construction is natural with respect to induced bundles, P determines a characteristic cohomology class depending only on the isomorphism class of the vector bundle. Since every characteristic class for complex vector bundles can be expressed as a polynomial in the Chern classes, the key remaining question is how the classes $(\sigma_r(K))$ relate to the Chern classes $c_r(\xi)$.

4 Real Vector Bundles and the Levi-Civita Connection

The concepts of connection and curvature are defined for real vector bundles in exactly the same way, replacing $\mathbb{C}$ by $\mathbb{R}$ throughout. Any invariant polynomial on $M_n(\mathbb{R})$ gives rise to a characteristic cohomology class in $H^*(M; \mathbb{R})$.

Let ξ be a real vector bundle equipped with a Euclidean metric, so that the inner product $\langle s, s' \rangle$ of two sections is a smooth real-valued function.

Definition 4.1 (Metric-compatible connection). A connection ∇ on ξ is *compatible with the metric* if

$$d\langle s, s' \rangle = \langle \nabla s, s' \rangle + \langle s, \nabla s' \rangle$$

for all sections s, s' . Here the inner products on the right are defined by the requirement that $\langle \alpha \otimes s, s' \rangle = \langle s, \alpha \otimes s' \rangle = \langle s, s' \rangle \alpha$ for all 1-forms α .

Lemma 4.2. Let $s_1, \dots, s_n$ be an orthonormal basis for the sections of $\xi|_U$, so that $\langle s_i, s_j \rangle = \delta_{ij}$. Then a connection ∇ on $\xi|_U$ is compatible with the metric if and only if the associated connection matrix $[\omega_{ij}]$ is skew-symmetric.

Now specialize to the dual tangent bundle τ^* of M .

Definition 4.3 (Symmetric / torsion-free connection). A connection ∇ on τ^* is *symmetric* (or *torsion-free*) if the composition

$$C^\infty(\tau^*) \xrightarrow{\nabla} C^\infty(\tau^* \otimes \tau^*) \longrightarrow C^\infty(\wedge^2 \tau^*)$$

is equal to the exterior derivative d . In terms of local coordinates, this means the Christoffel symbols are symmetric in their lower indices.

Lemma 4.4. The dual tangent bundle τ^* of a Riemannian manifold possesses one and only one symmetric connection which is compatible with its metric. This preferred connection is called the *Riemannian connection* or the *Levi-Civita connection*.

Sketch of proof: Apply the fact that any $n \times n \times n$ array of real valued functions A_{ijk} can be written uniquely as the sum of an array B_{ijk} which is symmetric in i, j and an array C_{ijk} which is skew-symmetric in j, k .

5 The Gauss–Bonnet Theorem for Surfaces

Let us specialize to a 2-dimensional oriented Riemannian manifold. With respect to an oriented local orthonormal basis θ_1, θ_2 for 1-forms, the connection and curvature matrices take the form

$$\omega = \begin{pmatrix} 0 & \omega_{12} \\ -\omega_{12} & 0 \end{pmatrix}, \quad \Omega = \begin{pmatrix} 0 & \Omega_{12} \\ -\Omega_{12} & 0 \end{pmatrix},$$

with $d\omega_{12} = \Omega_{12}$. The identity

$$\begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix} \begin{pmatrix} 0 & \Omega_{12} \\ -\Omega_{12} & 0 \end{pmatrix} \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} = \begin{pmatrix} 0 & \Omega_{12} \\ -\Omega_{12} & 0 \end{pmatrix}$$

shows that the exterior 2-form Ω_{12} is independent of the choice of oriented local basis, hence defines a global 2-form.

Definition 5.1 (Gauss–Bonnet 2-form). This form Ω_{12} is called the *Gauss–Bonnet 2-form* on the oriented surface. Denoting the oriented area 2-form $-\theta_1 \wedge \theta_2$ by dA , we can write $\Omega_{12} = K dA$, where K is a scalar function called the *Gaussian curvature*. Since both Ω_{12} and dA change sign if the orientation is reversed, K is independent of orientation.

In geodesic coordinates where the metric is $dx \otimes dx + g(x, y)^2 dy \otimes dy$, one finds $K = -g_{xx}/g$.

Theorem 5.2 (Gauss–Bonnet). For any closed oriented Riemannian 2-manifold M ,

$$\int_M \Omega_{12} = \int_M K dA = 2\pi e[M],$$

where $e[M]$ denotes the evaluation of the Euler class $e(\tau^*)$ on the fundamental homology class of M .

Sketch of proof:

1. An oriented 2-plane bundle ξ with Euclidean metric has a canonical complex structure J which rotates each fiber through $\pi/2$ in the counter-clockwise direction. In terms of an oriented local orthonormal basis s_1, s_2 , we have $J s_1 = s_2$.
2. For any compatible connection on ξ , the equations $\nabla s_1 = \omega_{12} \otimes s_2$ and $\nabla s_2 = -\omega_{12} \otimes s_1$ define a connection on the associated complex line bundle ζ (with J acting as multiplication by i). The connection matrix of this complex connection is $[i\omega_{12}]$ and the curvature matrix is $[i\Omega_{12}]$.
3. Applying the invariant polynomial $\sigma_1 = \text{trace}$ yields the closed 2-form $i\Omega_{12}$, which represents a characteristic cohomology class in $H^2(M; \mathbb{C})$. For complex line bundles, the only such class is the Chern class $c_1(\zeta) = e(\xi)$ and its multiples. Hence $(i\Omega_{12}) = \alpha e(\xi)$ for some complex constant α .
4. To determine α , take $\xi = \tau^*(M)$. Then $i \int_M \Omega_{12} = \alpha e[M]$. Evaluating on the unit 2-sphere S^2 (where $K \equiv 1$, $\text{area}(S^2) = 4\pi$, and $e[S^2] = 2$) gives $\alpha = 2\pi i$, as required.

6 Chern and Pontrjagin Classes in Terms of Curvature

The Gauss–Bonnet theorem for surfaces is a special case of a general theorem expressing the Chern classes of a complex vector bundle via curvature.

Theorem 6.1. Let ξ be a complex vector bundle with connection ∇ . Then the cohomology class $(\sigma_r(K))$ is equal to $(2\pi i)^r c_r(\xi)$.

To state this more compactly, define the invariant polynomial c by

$$c(A) = \det\left(I + \frac{A}{2\pi i}\right) = \sum_{k=0}^n \frac{\sigma_k(A)}{(2\pi i)^k}.$$

Then Theorem 6.1 is equivalent to the statement that the total Chern class satisfies

$$c(\xi) = (c(K)).$$

Proof: For a complex line bundle ζ , the Gauss–Bonnet computation gives $(\sigma_1(K)) = 2\pi i c_1(\zeta)$. Hence the cocycle $c(K) = 1 + \sigma_1(K)/2\pi i$ represents $c(\zeta) = 1 + c_1(\zeta)$.

If ζ splits as a Whitney sum $\zeta_1 \oplus \cdots \oplus \zeta_n$ of line bundles, choose connections on the ζ_j . The induced “Whitney sum” connection on ζ has diagonal curvature matrix $\Omega = \text{diag}(\Omega_1, \dots, \Omega_n)$, so $c(\Omega) = c(\Omega_1) \cdots c(\Omega_n)$. Thus $c(K) = c(K_1) \cdots c(K_n)$ represents $c(\zeta_1) \cdots c(\zeta_n) = c(\zeta)$.

The general case follows by a standard argument. If γ_1 denotes the universal line bundle over $\mathbb{P}^m(\mathbb{C})$ with m large, then the n -fold cross product $\gamma_1 \times \cdots \times \gamma_1$ satisfies the required identity. Since the cohomology of $\text{Gr}_n(\mathbb{C}^\infty)$ maps monomorphically into that of $(\mathbb{P}^m(\mathbb{C}))^n$ in dimensions $\leq 2m$, it follows that $c(\gamma_n) = (c(K(\gamma_n)))$. Therefore $c(\zeta) = (c(K))$ for an arbitrary bundle ζ .

Corollary 6.2. For any real vector bundle ξ , the de Rham cocycle $\sigma_{2k}(K)$ represents the cohomology class $(2\pi)^{2k}p_k(\xi)$ in $H^{4k}(M; \mathbb{R})$, while $\sigma_{2k+1}(K)$ is a coboundary. In other words, the total Pontrjagin class corresponds to the invariant polynomial

$$p(\xi) = \det\left(I + \frac{K}{2\pi}\right).$$

Remark 6.3. The vanishing of the odd-degree forms follows from the fact that for a metric-compatible connection, the curvature matrix is skew-symmetric, so $\sigma_m(\Omega) = \sigma_m(\Omega^t) = (-1)^m \sigma_m(\Omega)$.

Corollary 6.4. If a real (or complex) vector bundle possesses a flat connection, then all of its Pontrjagin (or Chern) classes with rational coefficients are zero.

7 The Euler Class and the Generalized Gauss–Bonnet Theorem

For an oriented $2n$ -plane bundle with Euclidean metric, one can express the Euler class in terms of curvature via the Pfaffian.

Lemma 7.1. There exists one and, up to sign, only one polynomial with integer coefficients which assigns, to each $2n \times 2n$ skew-symmetric matrix A over a commutative ring, a ring element $\text{Pf}(A)$ whose square is the determinant of A . Furthermore,

$$\text{Pf}(BAB^t) = \text{Pf}(A) \det(B)$$

for any $2n \times 2n$ matrix B . The sign is fixed by requiring $\text{Pf}(\text{diag}(S, \dots, S)) = +1$, where $S = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. This polynomial Pf is called the *Pfaffian*.

For example, $\text{Pf}\begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix} = a$, and for a 4×4 skew-symmetric matrix $[a_{ij}]$,

$$\text{Pf}(A) = a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23}.$$

Now let ξ be an oriented $2n$ -plane bundle with Euclidean metric. Choosing an oriented orthonormal basis for the sections over a coordinate neighborhood U , the curvature matrix $\Omega = [\Omega_{ij}]$ is skew-symmetric, so $\text{Pf}(\Omega)$ is a well-defined $2n$ -form on U . Under a change of oriented orthonormal basis, Ω is replaced by $X\Omega X^{-1}$ with X orthogonal and $\det X = 1$; since $X^{-1} = X^t$ and $\det X = 1$, the Pfaffian is unchanged. Thus the local forms piece together to a global $2n$ -form $\text{Pf}(K) \in C^\infty(\wedge^{2n}\tau^*)$. Just as before, one verifies that $d\text{Pf}(K) = 0$.

Since $\det(\Omega) = \text{Pf}(\Omega)^2$ represents $(2\pi)^{2n}p_n(\xi)$, the form $\text{Pf}(K)$ represents a class whose square is $(2\pi)^{2n}p_n(\xi) = (2\pi)^{2n}e(\xi)^2$. A sign check (using Whitney sums of 2-plane bundles) yields:

Theorem 7.2 (Generalized Gauss–Bonnet Theorem). For any oriented $2n$ -plane bundle ξ with Euclidean metric and any compatible connection, the exterior $2n$ -form $\text{Pf}(K/2\pi)$ represents the Euler class $e(\xi)$.

Remark 7.3. The Euler class *cannot* be expressed in terms of the curvature of an arbitrary (non-metric) connection. In fact, there exist oriented vector bundles with flat connections whose real Euler class is non-zero. This contrasts with the Chern and Pontrjagin classes, which vanish rationally for flat connections.